

# Adiar

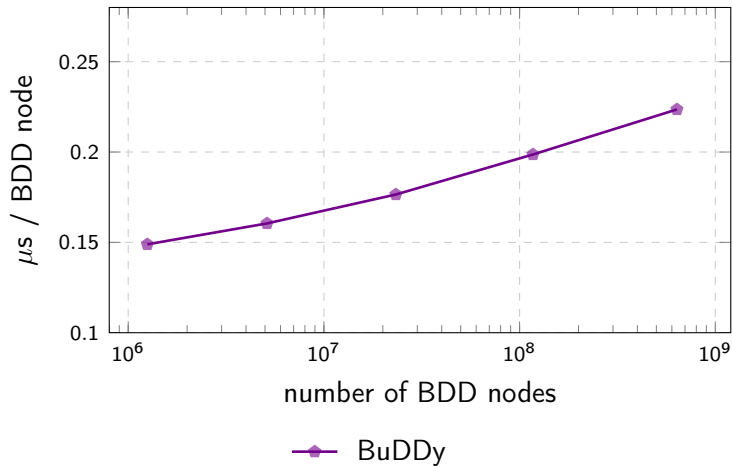
## Binary Decision Diagrams in External Memory

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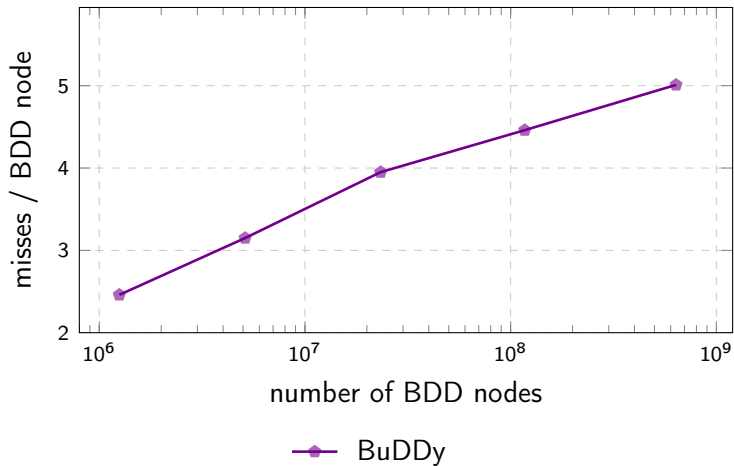
**Steffan Christ Sølvesten**, Jaco van de Pol,  
Anna Blume Jakobsen, and Mathias Weller Berg Thomasen  
TACAS 2022





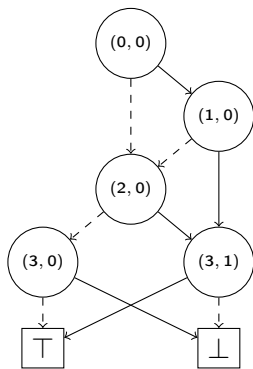


Minimal running time for the *Queens* problems.

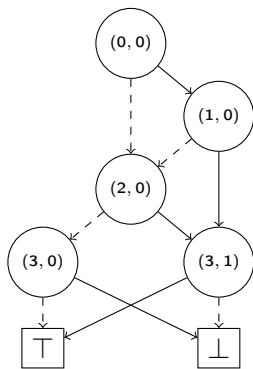


Cache-misses for the *Queens* problems.





**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

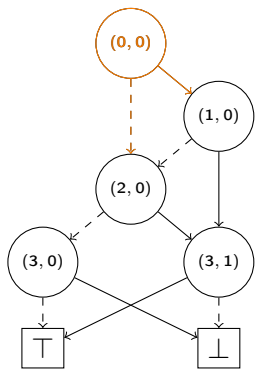


**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Priority Queue:  $Q_{count}$ :

[

]



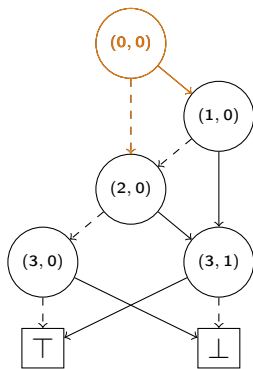
**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Priority Queue:  $Q_{count}$ :

[

]



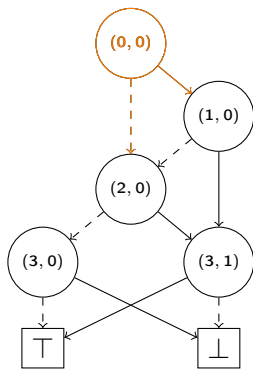


**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Priority Queue:  $Q_{count}$ :

[  $((0,0) \xrightarrow{T} (1,0), 1)$  ,  
 $((0,0) \xrightarrow{\perp} (2,0), 1)$  ,

]



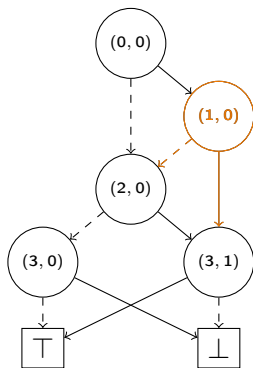
(a)  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek   | Sum | Result |
|--------|-----|--------|
| (1, 0) | 0   | 0      |

Priority Queue:  $Q_{count}$ :

[  $((0, 0) \xrightarrow{T} (1, 0), 1)$  ,  
 $((0, 0) \xrightarrow{\perp} (2, 0), 1)$  ,

]



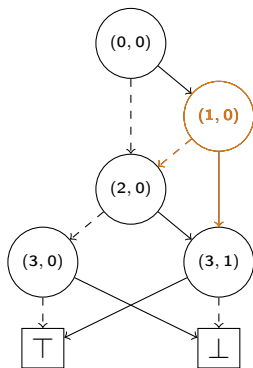
**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(1, 0)</b> | 0   | 0      |

Priority Queue:  $Q_{count}$ :

[  $((0, 0) \xrightarrow{\top} (1, 0), 1)$  ,  
 $((0, 0) \xrightarrow{\perp} (2, 0), 1)$  ,

]

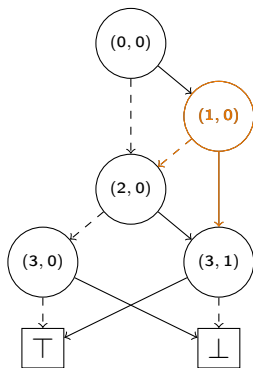


**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(1, 0)</b> | 1   | 0      |

Priority Queue:  $Q_{count}$ :

[  
 $((0, 0) \xrightarrow{\perp} (2, 0), 1)$  ,  
 ]

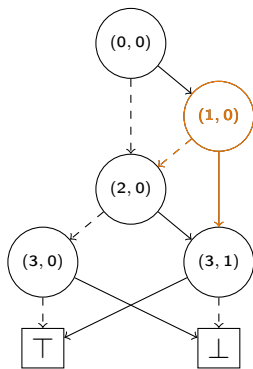


**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(1, 0)</b> | 1   | 0      |

Priority Queue:  $Q_{count}$ :

[  
 $((0, 0) \xrightarrow{\perp} (2, 0), 1)$  ,  
 $((1, 0) \xrightarrow{\perp} (2, 0), 1)$  ,  
 $((1, 0) \xrightarrow{\top} (3, 1), 1)$  ,  
 ]

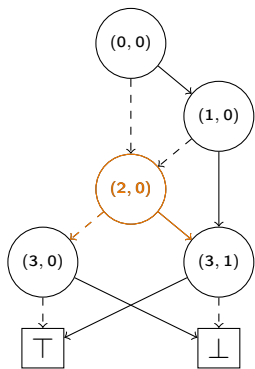


(a)  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek   | Sum | Result |
|--------|-----|--------|
| (2, 0) | 0   | 0      |

Priority Queue:  $Q_{count}$ :

[  
 $((0, 0) \xrightarrow{\perp} (2, 0), 1)$  ,  
 $((1, 0) \xrightarrow{\perp} (2, 0), 1)$  ,  
 $((1, 0) \xrightarrow{\top} (3, 1), 1)$  ,  
 ]

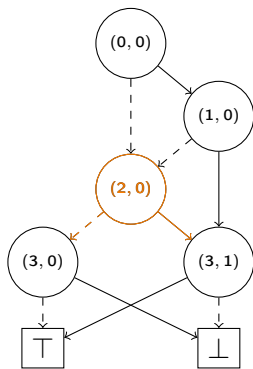


**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(2, 0)</b> | 0   | 0      |

Priority Queue:  $Q_{count}$ :

[  
 $((0, 0) \xrightarrow{\perp} (2, 0), 1)$  ,  
 $((1, 0) \xrightarrow{\perp} (2, 0), 1)$  ,  
 $((1, 0) \xrightarrow{\top} (3, 1), 1)$  ,  
 ]



**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(2, 0)</b> | 1   | 0      |

Priority Queue:  $Q_{count}$ :

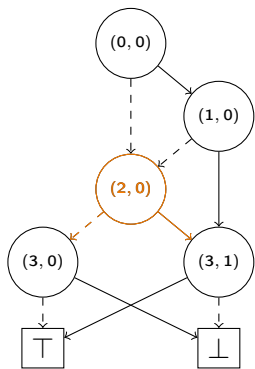
[

$((1, 0) \xrightarrow{\perp} (2, 0), 1)$  ,

$((1, 0) \xrightarrow{\top} (3, 1), 1)$  ,

]





**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

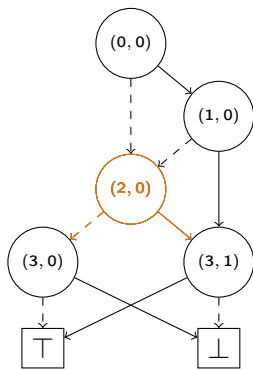
| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(2, 0)</b> | 2   | 0      |

Priority Queue:  $Q_{count}$ :

[

$((1, 0) \xrightarrow{T} (3, 1), 1)$  ,

]



**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(2, 0)</b> | 2   | 0      |

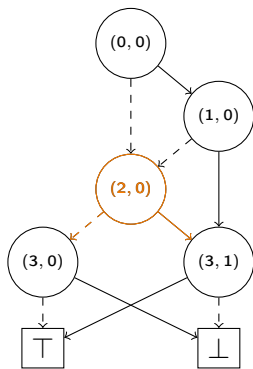
Priority Queue:  $Q_{count}$ :

[

$((2, 0) \xrightarrow{\perp} (3, 0), 2)$  ,

$((1, 0) \xrightarrow{\top} (3, 1), 1)$  ,

$((2, 0) \xrightarrow{\top} (3, 1), 2)$  ]



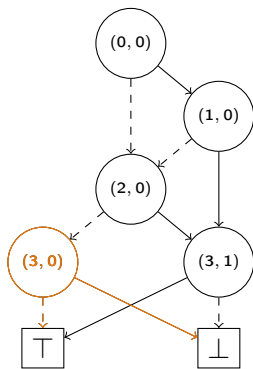
**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(3, 0)</b> | 0   | 0      |

Priority Queue:  $Q_{count}$ :

[

$((2, 0) \xrightarrow{\perp} (3, 0), 2)$  ,  
 $((1, 0) \xrightarrow{\top} (3, 1), 1)$  ,  
 $((2, 0) \xrightarrow{\top} (3, 1), 2)$  ]



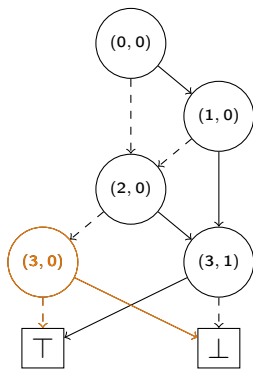
**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(3, 0)</b> | 0   | 0      |

Priority Queue:  $Q_{count}$ :

[

$((2, 0) \xrightarrow{\perp} (3, 0), 2)$  ,  
 $((1, 0) \xrightarrow{\top} (3, 1), 1)$  ,  
 $((2, 0) \xrightarrow{\top} (3, 1), 2)$  ]



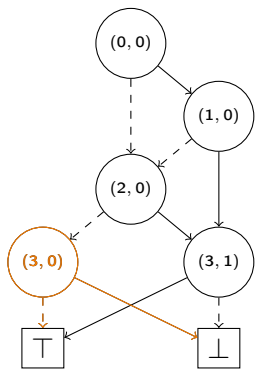
(a)  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek   | Sum | Result |
|--------|-----|--------|
| (3, 0) | 2   | 0      |

Priority Queue:  $Q_{count}$ :

[

$((1, 0) \xrightarrow{T} (3, 1), 1)$  ,  
 $((2, 0) \xrightarrow{T} (3, 1), 2)$  ]



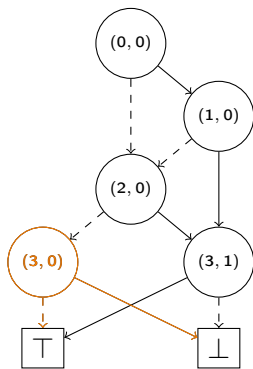
(a)  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek   | Sum | Result |
|--------|-----|--------|
| (3, 0) | 2   | 2      |

Priority Queue:  $Q_{count}$ :

[

$((1, 0) \xrightarrow{T} (3, 1), 1)$  ,  
 $((2, 0) \xrightarrow{T} (3, 1), 2)$  ]



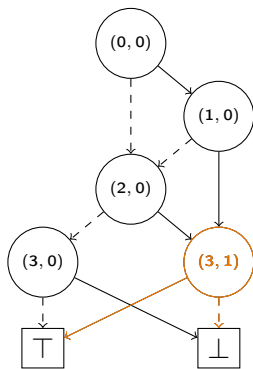
**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(3, 1)</b> | 0   | 2      |

Priority Queue:  $Q_{count}$ :

[

$((1, 0) \xrightarrow{T} (3, 1), 1)$  ,  
 $((2, 0) \xrightarrow{T} (3, 1), 2)$  ]



**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

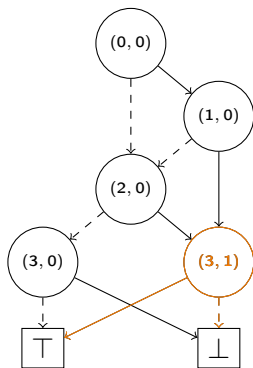
| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(3, 1)</b> | 0   | 2      |

Priority Queue:  $Q_{count}$ :

[

$((1, 0) \xrightarrow{T} (3, 1), 1)$  ,  
 $((2, 0) \xrightarrow{T} (3, 1), 2)$  ]





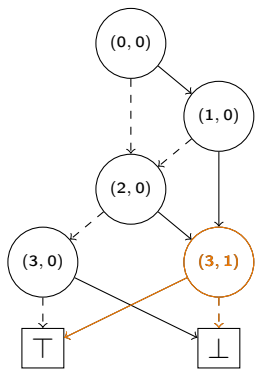
**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(3, 1)</b> | 1   | 2      |

Priority Queue:  $Q_{count}$ :

[

$((2, 0) \xrightarrow{T} (3, 1), \quad 2) \quad ]$



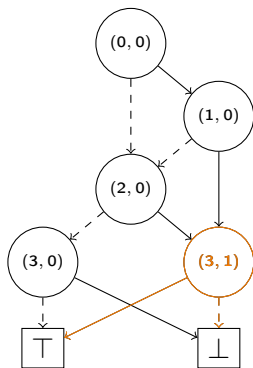
**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek          | Sum | Result |
|---------------|-----|--------|
| <b>(3, 1)</b> | 3   | 2      |

Priority Queue:  $Q_{count}$ :

[

]



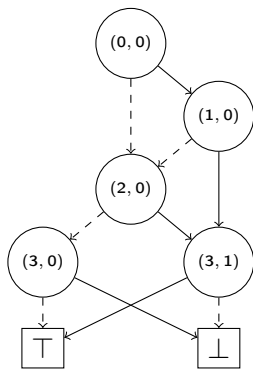
(a)  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

| Seek   | Sum | Result |
|--------|-----|--------|
| (3, 1) | 3   | 5      |

Priority Queue:  $Q_{count}$ :

[

]



**(a)**  $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Result

5

Priority Queue:  $Q_{count}$ :

[

]

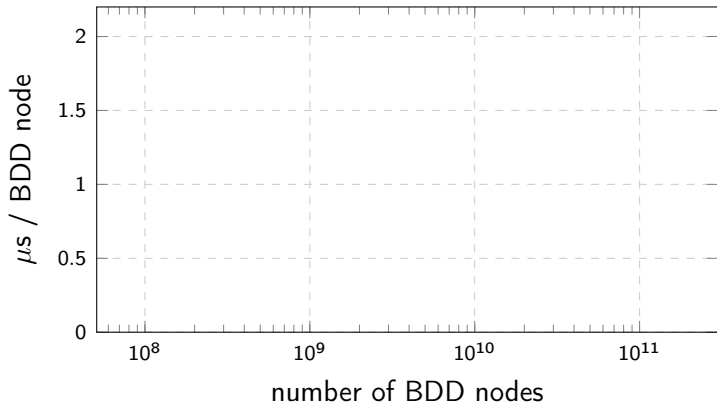


# Adiar

*I/O-efficient Decision Diagrams*

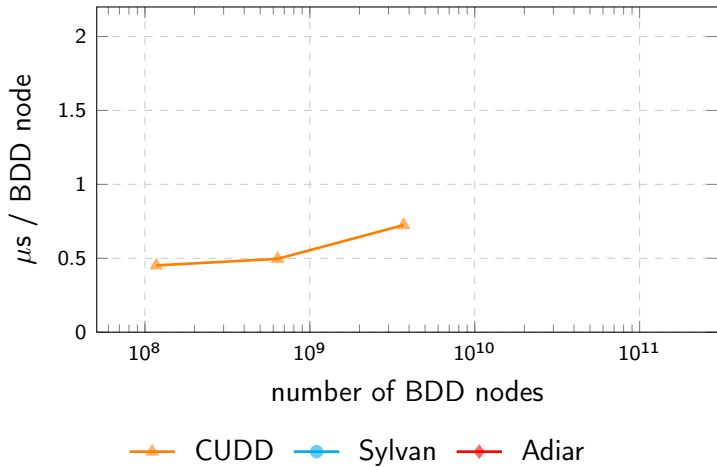
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[github.com/ssoelvsten/adiar](https://github.com/ssoelvsten/adiar)



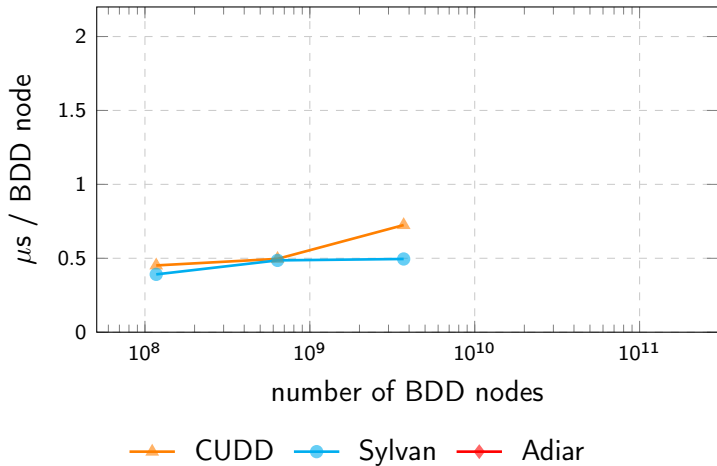
—▲— CUDD    —●— Sylvan    —◆— Adiar

Minimal running time for the *Queens* problems.

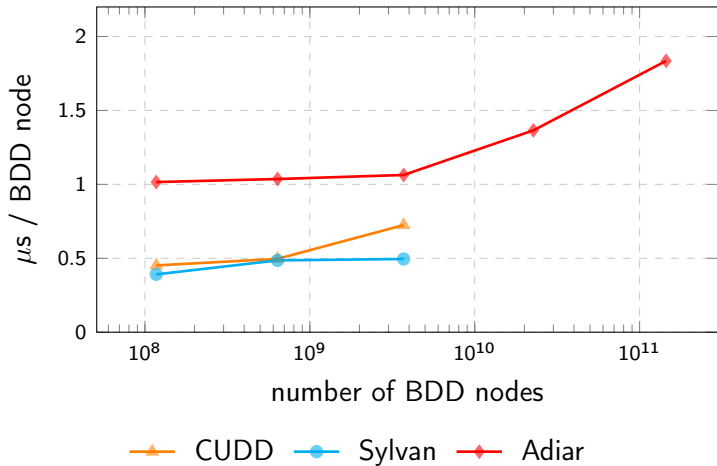


Minimal running time for the *Queens* problems.





Minimal running time for the *Queens* problems.



Minimal running time for the *Queens* problems.



| Algorithm                         | Time (s) |
|-----------------------------------|----------|
| $f \leftrightarrow g \equiv \top$ | 0.38     |

Checking the (EPFL Benchmark) *voter* circuit's single output gate ( $|N_f| = |N_g| = 5.76$  MiB).

| Algorithm                         | Time (s) |
|-----------------------------------|----------|
| $f \leftrightarrow g \equiv \top$ | 0.38     |
| $O(N \log N)$                     | 0.058    |

Checking the (EPFL Benchmark) *voter* circuit's single output gate ( $|N_f| = |N_g| = 5.76$  MiB).

| Algorithm                         | Time (s) |
|-----------------------------------|----------|
| $f \leftrightarrow g \equiv \top$ | 0.38     |
| $O(N \log N)$                     | 0.058    |
| $O(N)$                            | 0.006    |

Checking the (EPFL Benchmark) *voter* circuit's single output gate ( $|N_f| = |N_g| = 5.76$  MiB).



# Steffan Christ Sølvsten

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✉ [soelvsten@cs.au.dk](mailto:soelvsten@cs.au.dk)

🌐 [ssoelvsten.github.io](https://ssoelvsten.github.io)

## Adiar

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🔗 [github.com/ssoelvsten/adiar](https://github.com/ssoelvsten/adiar)

📖 [ssoelvsten.github.io/adiar](https://ssoelvsten.github.io/adiar)